Knowledge Distillation for Building Lightweight Models

UNIVERSITY OF TORONTO

Introduction

Why we use Knowledge Distillation?

- Increased Efficiency and Speed

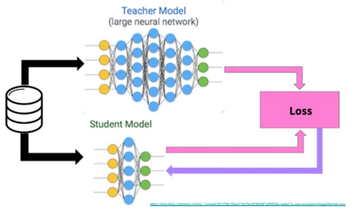
- Lower Energy Consumption

- Enhanced Accessibility

- Preservation of Model Performance

- Reduced Resource Requirements

What is Knowledge Distillation?



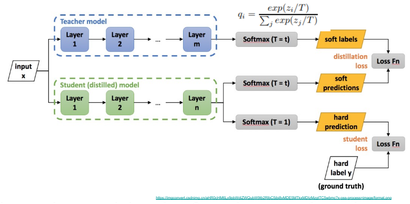
- Efficiently transferring knowledge from large/ complex model to small/efficient models

- Two main entities
  - High-capacity teacher model
  - Lower-capacity teacher model

- Challenge: Design an effective transfer mechanism that allows student capture key features while maintaining light weight advantage.

Methodology

Distilling the Knowledge in a Neural Network [1]



- Train teacher model and softened outputs by using a temperature variables in softmax function.

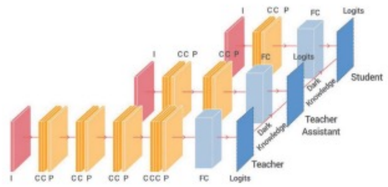
- Train student model replicate the hard output label and softened outputs of teacher model.

- When training student model first use same temperature in softening teacher model, after training back to 1.

- 2 loss function: one for difference between student prediction and train label (cross-entropy loss); second for difference between student and teacher's prediction (distillation loss).

State-of-the-art KD algorithm

- Knowledge Distillation via Teacher Assistant [2]



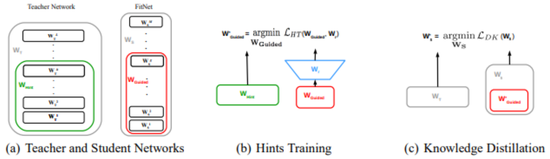
- When there is a large difference between the size of the student model and the teacher model, the performance of the student model decreases.

- The teacher model is used to distill the Teacher Assistant (TA) model, and then the TA model is used to distill the student model

- The TA model captures and distills the teacher's generalization abilities effectively, enhancing the student's performance over direct distillation.

- The method is particularly effective when the TA model's performance is close to the average of the teacher and student models.

- FitNets: hints for Thin Deep Nets [3]



- Deeper and thinner architectures for student networks

- Training:
  - Train teacher model and student model with hard label

  - Choose hint layer from teacher Networks and choose guided layer from student networks

  - Feature map size adjustment: A convolutional layer adapter (Wr) is used to align the student model's guided layer with the teacher's hint layer

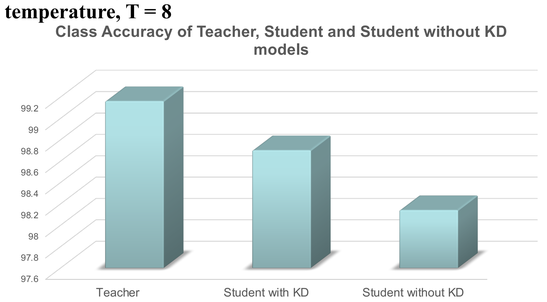
  - Optimize the student model with mean squared loss to align its guided layer's output closely with the teacher's hint layer.

  - Apply Knowledge Distillation (KD): The student model is distilled by training it to match the teacher's softened output probabilities, enhancing its generalization with the teacher's knowledge.

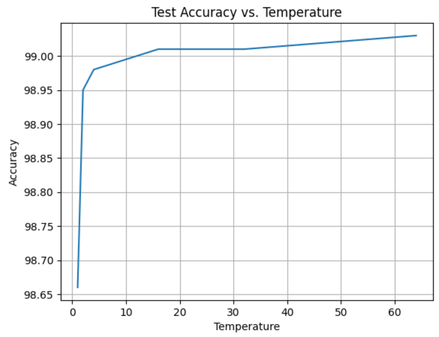
Results

- Knowledge Distillation from logits

Tuned Hyperparameters: **Task balance α = 0.7 and the distillation temperature, T = 8**

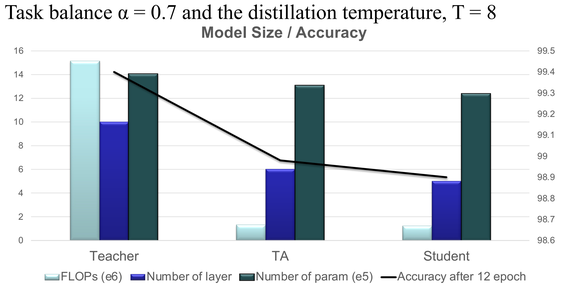


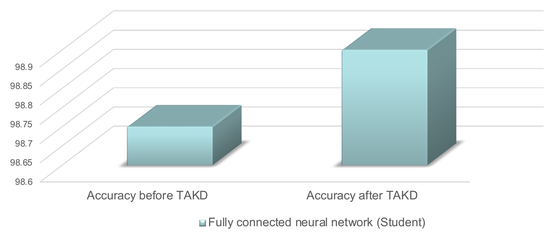
T = 1,2,4,16,32, and 64 when the task balance parameter is 0.5



- Knowledge Distillation via Teacher Assistant

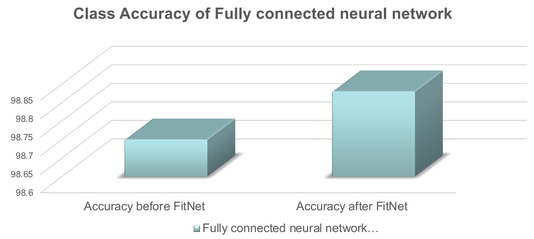
Task balance α = 0.7 and the distillation temperature, T = 8





- FitNets: hints for Thin Deep Nets

We chose:
Task balance **α = 0.9** and the distillation temperature, **T = 8**
hint_layer to be **teacher_model.layers[6]**
guided_layer to be **student_model.layers[2]**



Conclusions

- The method of using high-temperature soft targets, the introduction of a Teacher Assistant (TA) model in the knowledge distillation process, and FitNet all enhance the accuracy of the student model.

- Increasing temperature parameter T significantly improves test accuracy, especially in the range from T=1 to T=4.

- Indicating that the choice of task balance and temperature is crucial in the knowledge distillation process.

- An appropriately sized TA model effectively bridges the gap between teacher and student models, enhancing knowledge transfer efficiency.

- Underscoring the importance of choosing the right size for the TA model, as it must be sufficiently complex to capture the teacher's knowledge yet simple enough for the student to mimic.

- Choosing the appropriate intermediate layer as the hint layer is key in the FitNet approach and is a relatively complex process.

-FitNet highlights the significance of layer-specific knowledge transfer in deep neural networks.

Bibliography

1. **G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," arXiv.org, https://arxiv.org/abs/1503.02531 (accessed Oct. 30, 2023).**
2. **S. I. Mirzadeh et al., Improved knowledge distillation via teacher assistant, https://ojs.aaai.org/index.php/AAAI/article/view/5963/5819 (accessed Oct. 31, 2023).**
3. **A. Romero et al., "Fitnets: Hints for thin deep nets," arXiv.org, https://arxiv.org/abs/1412.6550 (accessed Oct. 30, 2023).**

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